

Sliding Window Level 1 Problems

1. Maximum Sum Subarray of Size K

→ Fixed Window

```
nums = [2, 1, 5, 1, 3, 2]
k = 3
```

```
def max_sum_subarr(nums, k):

    left = 0
    window_sum = 0
    max_sum = float('-inf')

    for right in range(len(nums)):
        window_sum += nums[right]

        if right - left + 1 == k:

            max_sum = max(max_sum, window_sum)

            window_sum -= nums[left]

            left += 1

    return max_sum

print(max_sum_subarr(nums,k))
```

2. Average of Subarray of Size K

→ Fixed Window

```
nums = [1, 12, -5, -6, 50, 3]
k = 4
```

```
def max_avg(nums,k):
    left = 0
    window_sum = 0
    max_sum = float('-inf')

    for right in range(len(nums)):

        window_sum += nums[right]

        if right - left + 1 == k:
            max_sum = max(max_sum,window_sum)
            window_sum -= nums[left]
            left +=1

    return max_sum / k

print(max_avg(nums,k))
```

3. Maximum Number of Vowels in a Substring of Size K

→ Fixed Window

```
s = "abciidef"
k = 3
```

```
def max_vowels(s,k):
    vowels = set('aeiou')
    left = 0
    max_cnt = 0
    vowels_cnt = 0

    for right in range(len(s)):

        if s[right] in vowels:
            vowels_cnt +=1

        if right - left + 1 == k:
            max_cnt = max(max_cnt,vowels_cnt)

            if s[left] in vowels:
                vowels_cnt -=1

            left +=1

    return f"Maximum Vowel is {max_cnt}"

print(max_vowels(s,k))
```